

Extension of specmine and WebSpecmine for Unsupervised Analysis

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Abstract. Metabolomics is a systems biology field focused on the comprehensive identification and quantification of metabolites in biological samples. specmine and WebSpecmine are bioinformatics tools developed by the BiSBI group at CEB, in collaboration with UFSC, for the integrated analysis of metabolomics and spectroscopy data. Despite their scientific impact, the current versions do not incorporate recent methodological advances in unsupervised learning, particularly techniques such as UMAP and t-SNE. This project proposes to extend these tools with new linear and non-linear dimensionality reduction methods and clustering algorithms, improving and validating both newly integrated and existing implementations using real-world case studies, and enhancing visualisation capabilities in WebSpecmine.

Keywords: Metabolomics · Unsupervised Learning · Clustering · UMAP · t-SNE · specmine · WebSpecmine

1 Introduction

Metabolomics is a systems biology field that aims at the comprehensive identification and quantification of the metabolites present in a biological sample. To this end, it relies on high-resolution analytical techniques such as Nuclear Magnetic Resonance (NMR), Gas Chromatography coupled to Mass Spectrometry (GC-MS), and Liquid Chromatography coupled to Mass Spectrometry (LC-MS), which enable the simultaneous measurement of hundreds or even thousands of metabolites. The result is high-dimensionality datasets whose manual analysis is unfeasible, making the use of specialised computational tools for their processing and interpretation indispensable.

Metabolomics has emerged as a key discipline in precision medicine and systems biology, enabling the identification of disease-associated biomarkers across a wide range of conditions, including cancer, diabetes, cardiovascular disorders, and neurodegenerative diseases [7]. Unlike genomics or transcriptomics, the metabolome is highly sensitive to both genetic and environmental factors, making it a direct readout of an organism’s physiological state. The growing integration of metabolomics with other omics layers — genomics, transcriptomics,

and proteomics — further amplifies its potential for understanding complex biological systems and supporting personalised therapeutic strategies [7].

The processing of these data involves multiple steps, from preprocessing (normalisation, alignment, noise removal) to the application of statistical and machine learning methods, with the goal of identifying relevant patterns, discriminating sample groups, and discovering potential biomarkers. In this context, the BiSBI group at CEB (Centre of Biological Engineering), in collaboration with the Federal University of Santa Catarina (UFSC, Brazil), developed bioinformatics tools dedicated to this analysis. Among these, the R package *specmine* stands out for the integrated analysis of metabolomics and spectroscopy data [1], alongside *WebSpecmine*, a web application built on *specmine* that allows users without programming experience to perform sophisticated analyses through an accessible graphical interface [2].

specmine provides a comprehensive set of methods covering data loading in multiple formats, preprocessing, metabolite identification, univariate and multivariate statistical analysis, machine learning, and variable selection. A fundamental advantage is its capacity to process different types of experimental data within a unified R environment. *WebSpecmine* complements this approach by supporting spectroscopic and chromatographic data (NMR, IR, UV-Vis, Raman, LC/GC-MS) and offering functionalities for multivariate analysis, metabolite identification, and metabolic pathway analysis, with user account management for data storage and sharing [2].

Unsupervised learning methods are fundamental in this pipeline and serve several key purposes: (i) drawing inferences about the intrinsic properties of the data, such as intra-cluster stability and the unsupervised of samples; (ii) assessing how preprocessing steps are affecting data structure, enabling quality control and method comparison; and (iii) preparing data for downstream supervised analyses — for instance, by mitigating overfitting through dimensionality reduction, removing outliers, and improving preprocessing decisions. Despite the maturity of *specmine* and *WebSpecmine*, their current implementations do not fully cover recent advances in unsupervised methodologies, motivating the extensions proposed in this project.

2 State of the Art

2.1 Existing Tools for Metabolomics Analysis

Several software platforms are currently available for metabolomics data analysis, each targeting different aspects of the analytical pipeline. Table 1 provides a comparative overview of the most widely used tools, highlighting key differences in scope, interface, and support for modern unsupervised learning methods.

XCMS Online is widely used for preprocessing of LC-MS and GC-MS data, offering feature detection, alignment, and statistical comparison, but focuses almost exclusively on preprocessing and does not include downstream machine learning methods [8]. MZmine provides flexible, modular workflows for mass

Table 1. Comparison of metabolomics analysis platforms.

Tool	Interface	Data Types	Clustering	PCA	UMAP/t-SNE
specmine	R package	NMR, MS, UV, IR, Raman	Yes	Yes	No
WebSpecmine	Web (GUI)	NMR, MS, UV, IR, Raman	Yes	Yes	No
MetaboAnalyst	Web (GUI)	NMR, MS	Yes	Yes	No
XCMS Online	Web (GUI)	LC/GC-MS	No	Yes	No
MZmine	Desktop	LC/GC-MS	No	Yes	No

spectrometry data processing but similarly lacks support for modern dimensionality reduction techniques [8]. MetaboAnalyst is arguably the most feature-complete web-based platform, offering preprocessing, statistical analysis, pathway analysis, and visualisation; however, it does not natively support UMAP or t-SNE [9]. specmine and WebSpecmine distinguish themselves by supporting multiple spectroscopic data types within a unified environment and by being fully open-source and extensible. As shown in Table 1, no existing platform currently integrates UMAP or t-SNE natively, representing a clear gap that this project addresses.

2.2 Unsupervised Learning in Metabolomics

Unsupervised learning encompasses a set of methods that identify patterns and structure in data without relying on predefined class labels. In metabolomics, these methods are applied in two main categories: **clustering** and **dimensionality reduction**.

Clustering. Clustering methods aim to group samples or variables according to their similarity, without prior knowledge of group membership. The main families of clustering algorithms relevant to metabolomics include:

- **Partitioning methods** (e.g., k-means, k-medoids): divide the dataset into a predefined number of k clusters by optimising an intra-cluster similarity criterion. k-means is frequently combined with PCA to identify discrete sample groups and has demonstrated strong reproducibility in survival prediction tasks based on metabolomics data.
- **Hierarchical methods** (e.g., agglomerative clustering, UPGMA): build a dendrogram of nested clusters, either bottom-up (agglomerative) or top-down (divisive), allowing visualisation of cluster structure at multiple resolutions.
- **Density-based methods** (e.g., DBSCAN, HDBSCAN): identify clusters as dense regions of points separated by sparser areas, with the ability to detect arbitrarily shaped clusters and label outliers as noise — a particularly useful property for metabolomics datasets where outliers may reflect sample degradation or analytical artefacts.

- **Model-based methods** (e.g., Gaussian Mixture Models): assume that data are generated from a mixture of probability distributions and estimate cluster assignments via Expectation-Maximisation, providing a probabilistic interpretation of cluster membership.
- **Spectral and kernel methods** (e.g., Spectral Clustering, SIMLR): use the eigenstructure of a similarity or graph Laplacian matrix to identify cluster structure, particularly effective for non-convex clusters. Comparisons across multiple unsupervised methods — including Sparse k-means, SIMLR, and Spectral Clustering — on metabolomics data have highlighted the utility of combining clustering with dimensionality reduction for biological discovery.

Dimensionality Reduction. Dimensionality reduction methods project high-dimensional data onto a lower-dimensional representation, preserving the most informative structure of the original data. They are divided into linear and non-linear approaches.

Linear Methods. Principal Component Analysis (PCA) remains the most widely used method, reducing data dimensionality by maximising explained variance and is already implemented in specmine [1]. PCA produces orthogonal components that are linear combinations of the original variables, making results interpretable but potentially insufficient when the relevant structure is non-linear. Other linear approaches such as Independent Component Analysis (ICA) and Factor Analysis are also applicable in metabolomics contexts.

Non-linear Methods. Non-linear methods, also known as manifold learning techniques, are capable of capturing complex, curved structures in high-dimensional data that linear projections cannot represent. Table 2 summarises the main methods considered in this project.

Table 2. Comparison of dimensionality reduction methods considered for integration in specmine.

Method Type		Preserves Global Structure	Scalability	In specmine
PCA	Linear	Yes	High	Yes
ICA	Linear	Partial	High	No
t-SNE	Non-linear	No	Low ($O(n^2)$)	No
UMAP	Non-linear	Yes	High	No

2.3 UMAP and t-SNE

Techniques such as UMAP (Uniform Manifold Approximation and Projection) and t-SNE (t-Distributed Stochastic Neighbor Embedding) have emerged as

powerful alternatives to classical dimensionality reduction approaches, having already demonstrated promising results in the analysis of high-dimensional omics data, namely in transcriptomics and metabolomics [3, 4]. Both methods model proximity relationships between points in high-dimensional space and project them onto a low-dimensional representation that preserves the local structure of the data. t-SNE is based on conditional probability distributions, while UMAP relies on fuzzy simplicial sets grounded in Riemannian geometry and algebraic topology [3].

Becht et al. [5] demonstrated that UMAP outperforms t-SNE in the visualisation of single-cell data, yielding more compact and consistent clusters across runs while better preserving global structure. Unlike t-SNE, UMAP simultaneously preserves local and global embedding structure and scales more efficiently with data size — its computational cost remains approximately linear in the number of samples, whereas t-SNE scales as $O(n^2)$ (or $O(n \log n)$ with the Barnes-Hut approximation). Kobak and Berens [6] further showed that many apparent advantages of UMAP over t-SNE can be largely attributed to differences in initialisation strategy, and proposed a protocol for more faithful t-SNE visualisations that includes PCA initialisation and adaptive learning rates — suggesting that both methods, when properly configured, remain highly relevant for omics data exploration. In applications to molecular dynamics and single-cell transcriptomics, UMAP has demonstrated greater cluster compactness, greater robustness to initialisation and data perturbation, and greater similarity to the original high-dimensional representation [5]. UMAP has also been applied in multi-omic integration, as demonstrated by the GAUDI framework, which uses UMAP embeddings to integrate and interpret multiple omic layers simultaneously.

This project therefore aims to extend specmine and WebSpecmine with new unsupervised learning methods, with emphasis on clustering algorithms and both linear and non-linear dimensionality reduction (notably UMAP and t-SNE), improving and validating both new and existing implementations using real-world case studies from UFSC, and enhancing the visualisation and interpretation capabilities available in WebSpecmine.

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